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# Formal Analysis of Lane-Changing Algorithms using Probabilistic Model Checking

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**Abstract**
Lane-changing algorithms play a critical role to ensure passenger safety and traffic efficiency in the dynamic and stochastic environment of Autonomous Vehicles (AVs). Despite the safety-critical nature of lane-changing algorithms, they are generally analyzed using computer simulation, which, due to its sampling-based nature, cannot guarantee capturing all the corner cases. As a more rigorous alternative, we advocate using probabilistic model checking for the formal analysis of lane-changing algorithms. The proposed approach utilizes Markov Decision Processes (MDPs) to model the stochastic dynamics of AV lane-changing maneuvers and allows us to formally verify properties specified in Probabilistic Computation Tree Logic (PCTL). For illustration, we formalized the MOBIL (Minimizing Overall Braking Induced by Lane Changes) algorithm with the Intelligent Driver Model (IDM), i.e., a widely used framework for AV lane changing, and formally verified its critical properties, such as safety and lane-change efficiency, temporal performance, and system robustness under dynamic traffic conditions using the probabilistic model checker PRISM.

**Keywords** Formal analysis · Probabilistic analysis · Lane-changing algorithms · Formal verification · MOBIL · Autonomous vehicles

## 1 Introduction

Autonomous Vehicles (AVs) represent a transformative advancement in modern transportation technology, with profound potential to enhance safety, efficiency, and accessibility on a global scale. These systems use sensors, machine learning algorithms, and sophisticated control mechanisms to move around in the complex, real-world surroundings with minimal human involvement [1]. Central to the operational efficacy of AVs is the process of path planning [2]—a computational process by which an AV determines the optimal route and corresponding maneuvers required to progress from its current location to a designated destination.

Lane changing [3] is a crucial and one of the most complex tasks of path planning, and its effect is substantial on the safety and efficiency of roadway systems. Because of the dynamic, sometimes stochastic nature of road environments, it is very important for AVs to make lane-changing decisions efficiently, almost optimally, in real-time to avoid collisions. Lane changing involves transitioning between lanes to optimize traffic flow, maintain safety, or adhere to traffic regulations while interacting with other vehicles in shared roadways.

Lane-changing algorithms [3] have to successfully contend with the intricacies of the shared roadways that pose autonomous and human-driven vehicles with varying degrees of predictability. The imperative for safe and effective lane-changing is further underscored by empirical evidence from real-world incidents. According to the National Highway Traffic Safety Administration (NHTSA) documentation, all 11 crashes of Tesla models on autopilot system reveal potential challenges in autonomous systems in

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terms of path planning and lane changing [4]. Also, NHTSA predicts that there will be thousands of crashes with Level 2 [5] AVs (which require driver supervision) and about 200 attended incidents involving higher-level AVs (with some driver interventions to full automation) per year. These projections make one case for rigorous verification frameworks that ensure AVs execute as intended, perform the desired behaviors, and fulfill their safety constraints in their decision making. Robust lane-changing algorithms not only reduce the risk of vehicular accidents but also offer broader societal benefits, including alleviating traffic congestion, decreasing emissions, and enhancing the overall reliability of transportation infrastructure.

The verification of lane-changing algorithms for AVs has historically relied on computer simulations, which generally compromise the rigor of the analysis to make it computationally feasible and, thus, often overlook rare but critical corner cases. Correctness and reliability assessment of AV systems are enormously hard problems. AVs operate within highly unpredictable environments, making it practically infeasible to generate test cases that cover the full spectrum of possible scenarios. Prior studies underscore the magnitude of this challenge: To demonstrate, with 95% confidence, that the rate of failure of AVs is 20% less than for human drivers (1.09 fatalities per 100 million miles, US, 2013), even despite an AV fleet of 100 vehicles operating 24/7, 365 days/year at an average speed of 25 mph, would take around 400 years [6]. Such a level of empirical testing is practically impossible to provide safety guarantees. In a further example, back in 2007, CalTech developed an AV known as Alice for DARPA Urban Challenge [7]. Alice's goal was to navigate through an urban environment, where its tasks included parking and obeying traffic regulations. Nonetheless, during the competition, Alice showed unsafe behavior, almost *'owning'* a collision. Subsequently, the root cause was found in the adverse interaction between the reactive obstacle avoidance subsystem and the reacting path planner. The failure was in a very specific set of circumstances, and even with a lot of testing, it would have been very hard to notice on something that would only happen in one out of a few million [8].

Formal methods [9] have the potential to cater for the challenges mentioned above. However, the environmental uncertainties, the unpredictable traffic patterns, and obstacle presence, in the case of lane changing pose substantial challenges for traditional formal verification methods. To capture these uncertainties and unpredictable aspects, we propose a probabilistic model checking-based approach to verify the safety and performance of lane-changing algorithms. Probabilistic model checking [10] entails constructing mathematical representations, such as Markov Decision Processes (MDPs) [11], to capture the probabilistic dynamics of the system. MDPs allow decision-making within an uncertain environment that is captured through probabilistic transitions and multiple decision pathways. Probabilistic model checking also facilitates a comprehensive verification process involving probabilistic properties expressed in Probabilistic Computation Tree Logic (PCTL) [12]. The main contribution of this paper is a generic modeling and verification approach for analyzing lane-changing algorithms. In this regard, we propose MDP-based generic models of sub-modules of the lane-changing process. These models can, in turn, be used to readily develop a formal model of any lane-changing algorithm and verify it using probabilistic model checking.

To illustrate the practical effectiveness of the proposed probabilistic model checking based approach for analyzing AV path planning, we formally analyze the Minimizing Overall Braking Induced by Lane Changes (MOBIL) model [13], i.e., an influential decision-making framework for lane-changing maneuvers in autonomous driving. The MOBIL model is meant to avoid flow disturbances like excessive braking while considering both safety and comfort concerns. Despite the fact that the practical utility of MOBIL has been shown in several simulation environments, to the best of our knowledge, its rigorous formal verification is still lacking. Utilizing the proposed approach, we formally modeled the MOBIL model coupled with the Intelligent Driver Model (IDM), a widely-used car-following model that computes longitudinal accelerations, using MDPs. This enables a rigorous evaluation using a probabilistic model checker of its decision-making process under varying traffic conditions, uncertainties, and potential failures. For this evaluation, we employ the PRISM [14] probabilistic model checker, which supports the development and analysis of MDP-based formal models.

This paper advances the formal analysis of lane–changing with a general workflow and a case study. The main contributions are as follows.

- We propose a systematic methodology that decouples lane-change decision logic from longitudinal dynamics for formal verification of various algorithms via probabilistic model checking. High-level decision rules are translated into formal acceleration guards, forming a closed-loop MDP analyzed using PCTL.
- We verify PCTL and reward properties under min–max semantics, providing scheduler-robust bounds and reachability guarantees for safety, liveness, and performance.
- We define the lane-change rate with a reward ratio that provides exact counts and rates, giving clear semantics and enabling reproducible cross-strategy comparisons.

- Instantiating the framework with the MOBIL and the IDM, we identify a critical follower braking parameter $b_{\text{safe}}$, defined as the maximum acceptable deceleration (in $\text{m}/\text{s}^2$) for a following vehicle after a lane change (Definition 17). When $b_{\text{safe}} \leq -4\ \text{m}/\text{s}^2$, which is consistent with IDM's comfortable braking parameter $b$, we formally certify that critical unsafe states are provably unreachable, with minimum reachability probability equal to zero under all possible executions and scheduling strategies. For the same safety threshold, we additionally verify high-probability recovery from critical states within ten steps with at least $90\%$ probability, ensuring that the system returns to safe operation. Finally, we reproduce the canonical rate-versus-density shape while quantifying trade-offs over politeness factor $p$ and incentive threshold $a_{\text{thr}}$.

The remainder of the paper is organized as follows: Section 2 reviews the related work in the field. Section 3 describes the preliminaries to facilitate the understanding of the underlying concepts. Section 4 presents the proposed approach, and Section 5 proposes generic formalizations of lane-changing algorithms, which in turn serve as the basis for formally verifying the MOBIL model discussed in Section 6. Section 7 wraps up this paper by summarizing the findings and mentioning some directions for future work.

## 2 Related Work

Lane changing has garnered significant scholarly attention due to its critical role in ensuring safety, efficiency, and reliability for AVs. Traditional lane-changing methodologies encompass a range of graph-based algorithms, such as Dijkstra's algorithm [15] and A* [16]. They proved to be very effective in the static or moderately changing environment where the positions of the obstacles and road participants are easy to predict. The algorithms that derive from this idea have been fundamental tools for computing shortest paths, especially in static environments, because they allow efficient shortest path computation. However, they can often be insufficient in highly dynamic environments that introduce rapid changes of traffic elements which go beyond the scope of static planning paradigms. To overcome these shortcomings, sampling-based methods including the Probabilistic Roadmaps (PRM) [17] and Rapidly-exploring Random Trees (RRT) [18] have received more attention lately. These methods incorporated stochasticity to cope with the dynamic nature of complex environments. The benefit of generating feasible trajectories for both holonomic and non-holonomic systems is provided. In addition, the RRT* [19] can even guarantee the asymptotic optimality of the generated path, formally ensuring the almost-sure convergence to global optimal solutions, with the increasing number of samples. The sampling-based methods and their variants have been widely applied [15, 18] to AVs. The computational complexity of the sampling procedure, however, is high, making their practical use limited.

Machine learning, particularly using reinforcement learning, is being increasingly used to improve decision-making in uncertain and dynamic environments. For example, Yang et al. [20] presented a model that combines Long Short-Term Memory (LSTM) networks with Deep Deterministic Policy Gradients (DDPG) to enhance trajectory prediction in lane changes. By incorporating temporal learning into AV decision-making, they improved average single-step rewards by 7.4% over traditional methods. Furthermore, hybrid frameworks that integrate reinforcement learning with traditional rule-based methods were introduced to benefit from the adaptability of learning-based techniques while keeping the interpretability of deterministic models. A framework combining Deep Q-learning with rule-based constraint was introduced by Ghimire et al. [21], resulting in a safety rate of 0.8 while ensuring improved decision-making efficiency. Tian et al. [22] used personalized lane change assistance systems based on mode predictive control in conjunction with end-to-end imitation learning. This approach is adapted to the particular driver's behavior, giving it a driver-tailored driving experience, which is useful for human like AV behavior modeling. Moreover, Shi et al. [23] proposed a method to use imitation learning as reinforcement learning to initialize learned lane-changing behaviors, which improve collision rates and speed performances considerably.

Formal methods have also been used to analyze lane-changing algorithms. For example, Zita et al. [24] formally analyzed the lane-changing module of an AV, finding defects both in the model and its implementation. These results demonstrate the ability of formal methods to improve software reliability for autonomous systems. Similarly, Yang et al. [25] applied probabilistic model checking to multimodal transportation systems, optimizing path planning by dynamically adjusting congestion probabilities based on IoT sensor data. Dhonthi et al. [26] have also applied Signal Temporal Logic (STL) to verify the safety and the efficiency of AV path planning. Given temporal logic specifications, they verified AV trajectories in scenarios that may arise in real time (e.g., automated valet parking). This approach assures that safety standards are fulfilled by varying constraints that demonstrate that the AV behaves appropriately to static and dynamic obstacles. Chen and Li [27] also suggested a hybrid approach that combines data-driven

and model-driven approaches to advance formal modeling and analysis of the driving scenarios. They abstract complex models and map them to formal structures to provide a mechanism that improves the safety analysis of AV path planning.

Micro-simulation studies couple lane-change rules to realistic longitudinal dynamics and reveal robust qualitative trends, such as a peak in lane-change rate at intermediate density and a lower peak for higher politeness. Data-driven and reinforcement learning approaches optimize comfort and efficiency for particular traffic models. These works are strong on realism and on coverage of scenarios. They are weaker on guarantees and on comparability. First, safety is observed and not proven, so rare corner cases may be missed. Second, tie-breaking and scheduler choices can change outcomes but are rarely examined. Third, reported rates are often sensitive to sampling and binning and lack a common formal definition. Our framework addresses these gaps. We verify closed-loop PCTL properties that give min–max bounds and reachability results. We define density-conditioned rewards that induce rate curves with precise semantics. In this paper we instantiate the framework with a MOBIL lane-change strategy and an IDM longitudinal dynamics model. The closed-loop coupling maintains the qualitative relationship between lane-change rate and traffic density observed in previous studies. Additionally, when employing recommended braking thresholds, it provides a formal assurance that critical states are unattainable within our abstraction. These results complement micro-simulation by adding certified safety and reproducible measurement definitions while keeping realistic longitudinal behavior.

## 3 Preliminaries

We introduce the key concepts and tools needed for the formal analysis of lane-changing algorithms to provide the necessary background for the methods used in this paper.

### 3.1 Markov Decision Processes (MDPs)

MDPs allow modeling systems that exhibit both nondeterministic and probabilistic behavior. That's why we use MDPs to model the ego vehicle, surrounding vehicles, and lane-changing algorithms.

**Definition 1** (MDP) An MDP is defined as tuple $\mathcal{M} = (\mathcal{S}, s', \mathcal{A}, \mathcal{P}, AP, L)$, where:

- $\mathcal{S}$ is finite set of states;
- $s' \in \mathcal{S}$ is the initial state;
- $\mathcal{A}$ is a finite set of actions;
- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is a probabilistic transition function, where $\forall s \in \mathcal{S}, a \in \mathcal{A} : \sum_{s' \in \mathcal{S}} \mathcal{P}(s, a, s') \in \{0, 1\}$;
- AP is a set of atomic propositions;
- $L : \mathcal{S} \to 2^{AP}$ is a labeling function, such that $q \in L(s)$ if and only if $q$ is true in $s \in \mathcal{S}$.

In each state $s$ of an MDP $\mathcal{M}$, a choice is made between the actions that are enabled in $s$. These actions form the set $\mathcal{A}_s = \{a \in \mathcal{A} \mid \mathcal{P}(s, a, s') > 0 \ for\ some\ s \in \mathcal{S}\}$. When an action $a \in \mathcal{A}_s$ is selected in state $s$, the probability of transitioning to the next state $s'$ is give by $\mathcal{P}(s, a, s')$. A sequence of such transitions, denoted $\sigma = s_0 \xrightarrow{a_0} s_1 \xrightarrow{a_1} s_2 \ldots$, where $\mathcal{P}(s_i, a_i, s_{i+1}) > 0$ for $i \in \mathbb{N}$, represents an (infinite) path through the MDP. A finite path $\rho = s_0 \xrightarrow{a_0} s_1 \xrightarrow{a_1} \ldots \xrightarrow{a_{n-1}} s_n$ is defined as prefix of an infinite paths. The set of all finite and infinite paths of $\mathcal{M}$ starting from state $s$ are denoted by $\mathrm{FPath}_{\mathcal{M},s}$ and $\mathrm{IPath}_{\mathcal{M},s}$, respectively. The choice of action to take at each step of the execution of an MDP $\mathcal{M}$ is made by a policy, which makes its decision based on the history of $\mathcal{M}$ up to the current state.

**Definition 2** (Policy) A policy for MDP $\mathcal{M}$ is a function $\pi : \mathrm{FPath}_{\mathcal{M},s'} \to \mathcal{A}$ such that, for any path $\rho$ ending in state $s_n$, we have $\pi(\rho) \in \mathcal{A}_{s_n}$.

For a given fixed policy $\pi$, the nondeterminism of $\mathcal{M}$ is eliminated, and the system becomes fully probabilistic. More specifically, this corresponds to an infinite discrete-time Markov chain (DTMC), which enables the analysis of probabilistic behaviors over the infinite paths of $\mathcal{M}$.

**Definition 3** (Reward Structure) A reward structure in the context of an MDP $\mathcal{M}$, is formally defined as a function $\mathcal{R} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}_{\geq 0}$.

Of particular interest in this paper is the expected amount of reward accumulated up until a target is reached.

**Definition 4** (Expected Cumulative Reward) For a reward structure $\mathcal{R}$ on an MDP $\mathcal{M}$ and a target label $b \in AP$, we define the function $cumul^b_{\mathcal{R}} : \mathsf{IPath}_{\mathcal{M},\mathsf{s}} \to \mathbb{R}_{\geq 0}$ as:

$$cumul^b_{\mathcal{R}}(s_0 \xrightarrow{a_0} s_1 \xrightarrow{a_1} \ldots) = \sum_{i=0}^{n_b - 1} \mathcal{R}(s_i, a_i, s_{i+1}),$$

where $n_b$ is the first index for which $b \in L(s_{n_b})$. For the cases where $b \notin L(s_i)\ \forall i$, we define $n_b = \infty$.
Now, the maximum expected value for $cumul^b_{\mathcal{R}}$ is given by:

$$E^{max}_{\mathcal{M},s}(cummul^b_{\mathcal{R}})$$

For a cost structure $c : \mathcal{S} \times \mathcal{A} \to \mathbb{R}_{\geq 0}$, the expected cumulative cost problem maintains a similar definition after substituting max with min in the above-mentioned expression.

### 3.2 PCTL

PCTL [12] is the extension of Computation Tree Logic (CTL) [28], which allows reasoning about infinite sequences of states.

**Definition 5** (Syntax) A state formula $\varphi$ is defined as:

$$\varphi :: = \text{true} \mid p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \text{X}\varphi \mid P_{\sim q}(\Box\varphi) \mid P_{\sim q}(\varphi \text{U} \varphi),$$

where $p$ is an atomic proposition ($p \in AP$), $\sim$ is a comparison operator ($\sim \in \{<, \leq, >, \geq\}$), $q$ is a probability threshold ($q \in [0, 1]$), and $\varphi$ is a path formula. The propositional connectives have the usual meaning. The X operator provides the meaning "next state," thus indicating the target formula will hold true during the subsequent state. The U operator is read "until," meaning that its second argument will eventually become true in some state, and the first argument will be continuously true until this point.

### 3.3 Property Specifications

MDPs can be checked for two property categories which include probabilistic reachability alongside expected reachability evaluation. A state predicate together with these properties determines the probability of achieving states that satisfy the predicate as well as the reward expectation before reaching them. Property values depend on how well nondeterministic choices (i.e., policy) are resolved. All possible strategies are examined when we seek optimal (minimum or maximum) values. The "min" usage is a matter of whether we are searching for adversarial conditions or a "best possible" strategy.

### 3.4 PRISM Model Checker

The probabilistic model checker PRISM [14] is a widely used probabilistic model checker designed for analyzing various probabilistic models, including MDPs. PRISM provides a high-level modeling language, i.e., the PRISM language, based on the Reactive Modules formalism for constructing and analyzing complex systems [29]. In PRISM, models are described as a system of interacting modules where a module consists of a finite set of variables, which, taken as a whole, describes the state of the corresponding module. The transitions of a module are specified through a series of guarded commands in the following format:

`[action] <guard>→<prob>:<update> + ⋯ + <prob>:<update>`

A command does contain a guard, and optionally an action label, and a probabilistic choice between updates. A guard is a logical predicate over variables, which allows command execution only when it is true. The system moves to a new state according to the specified updates with probabilities given by `<prob>`. Actions facilitate the interaction between modules by allowing synchronization and communication between several components.

Support for reward structures in PRISM is provided through reward commands, defined as follows:

`[action]<guard>:<reward>`

representing the rewards accumulated when taking an action in a state that is satisfying the guard.

Since PRISM explores all reachable states systematically, it can compute probabilities, expected cumulative rewards, and so on, for a variety of properties—yielding rigorous results that are complementary to, and more exhaustive than, pure simulation.

## 4 Proposed Approach

Our proposed approach, illustrated in Fig. 1, provides a structured methodology for modeling, analyzing, and verifying lane-changing algorithms in AVs. It begins by defining key parameters of the ego vehicle (e.g., position, speed, acceleration, and lane) and the surrounding vehicles (e.g., presence, speed, and distance), ensuring a comprehensive representation of the traffic environment. These parameters are used to build an MDP, capturing the probabilistic nature of the vehicles' interactions and the autonomous decision-making process. Decision-making algorithms–including rule-based, reinforcement learning, or widely-studied approaches like MOBIL–are then integrated and evaluated against requirements specified in temporal logic. Leveraging a probabilistic model checker such as PRISM, the framework systematically generates quantitative performance and safety metrics, providing a sound basis for both verifying the correctness of lane-changing strategies and guiding their iterative improvement.

## 5 Formal Framework for Lane Change Analysis

In this section, we present fundamental definitions, lemmas, and propositions that abstractly describe the state space, transition dynamics, incentive and safety criteria, and verification properties of autonomous lane-changing scenarios. These formal constructs provide a flexible,

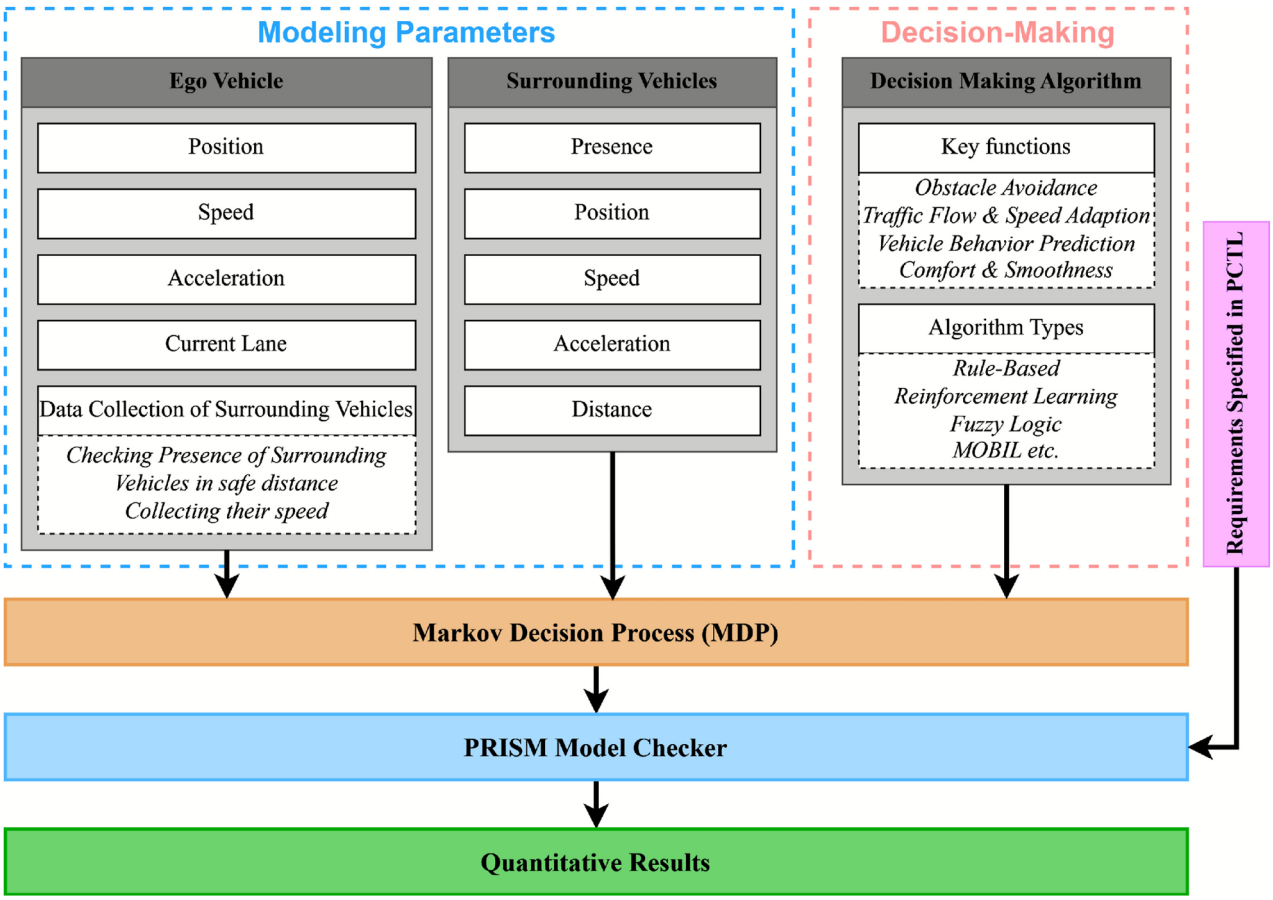



**Fig. 1** Proposed framework

algorithm-agnostic template that can be instantiated for a wide variety of decision-making strategies of lane-changing algorithms.

## 5.1 Formal Definitions and Theoretical Constructs

To formalize the modeling of a roadway environment for AVs, we first define the scenario configuration, which outlines the basic structure and components of the roadway.

**Definition 6** (Scenario Configuration) Consider a roadway with a fixed number of lanes $L >= 1$. An autonomous ego vehicle and multiple other vehicles are situated on this roadway. Each vehicle occupies a position in a discrete set of reference points along and cross the lanes. The state of each vehicle is described at discrete time steps.

The state of the autonomous ego vehicle is characterized by key parameters that describe its operational status, position, and speed within the roadway environment.

**Definition 7** (Ego Vehicle State) An ego vehicle is characterized by a state tuple $(s, l, v)$, where: $s \in \mathcal{S}$ is the operational state of the vehicle (e.g., cruising, changing lanes, keeping lane), $l \in \{1, 2, \cdots, N\}$ denotes the current lane number, with $N$ being the total number of lanes and $v \in \mathcal{V}$ represents the speed levels of the vehicle, where $\mathcal{V}$ is a finite set of possible speeds.

Similarly, to account for interactions with other vehicles, we define the surrounding vehicle states, which describe the presence, speed, and distance of other vehicles relative to the ego vehicle.

**Definition 8** (Surrounding Vehicles) For each position relative to the ego vehicle (e.g., ahead in the current lane, behind in right lane), we define: $presence_i \in \{true, false\}$ is a Boolean variable indicating the presence of a vehicle at position $i$, $speed_i \in \mathcal{V}$ is the speed of the vehicle at position $i$, if present, and $distance_i \in \mathcal{D}$ is the distance of vehicle at

position $i$ from ego vehicle, where $\mathcal{D}$ is a finite set of possible distances.

**Definition 9** (State Space) Let $\mathcal{S}$ be the finite state space capturing all the relevant system aspects. Each state $s \in \mathcal{S}$ is defined by tuple:

$$s = (Ego\ Vehicle\ State, \{presence_i, speed_i, distance_i\}_{i \in \mathcal{J}})$$

where *Ego Vehicle State* is as in Definition 7, and $\mathcal{J}$ is the index set of all defined surrounding positions. For each position $i \in \mathcal{J}$, if $presence_i = false$, we take $speed_i = 0$ by convention (meaning no vehicle is present at *i*), and if $presence_i = true$, then $speed_i \in \mathcal{V}$ is the (discretized) speed of that vehicle and $distance_i \in \mathcal{D}$ is its distance from the ego vehicle.

**Definition 10** (Transition Model) The system evolves based on a probabilistic transition model: $\mathcal{T} : \mathcal{S} \times \mathcal{A} \to \mathcal{P}(\mathcal{S})$, where $\mathcal{P}(\mathcal{S})$ is the probability distribution over the global state space. For a given state $s$ and action $a$, $\mathcal{T}(s, a)(s')$ represents the probability of transitioning to state $s'$.

**Definition 11** (Utility Function) The ego vehicle is associated with a utility function $\mathcal{U}_{\text{ego}} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ capturing the preferences from various outcomes. The utility function may consider safety, efficiency, and comfort.

In our context, $\mathcal{U}_{\text{ego}}$ may serve as the reward function $\mathcal{R}$ as in Definition 3 for the MDP model (or it can be derived from $\mathcal{R}$ by combining different reward components). The goal of a lane-changing algorithm is typically to maximize the cumulative utility (or minimize cost) while satisfying all safety constraints.

**Definition 12** (Feasibility of Lane Change) A lane change for the ego vehicle from lane $l$ to $l'$ (where $l' \in \{l-1, l+1\}$) is feasible if:

1. $l'$ is within the set of lanes $\mathcal{L}$, and
2. The lateral gap in lane $l'$ is sufficient to accommodate the ego vehicle i.e., $\forall i \in \mathcal{J}_{l'}$, if $presence_i = true$, then $distance_i \geq d_{min}$

Where $\mathcal{J}_{l'}$ is the set of positions (e.g., ahead, behind) relative to ego vehicle in lane $l'$, $distance_i$ is the distance between ego vehicle and vehicle at position $i$ and $d_{min}$ is safety threshold ensuring sufficient clearance for lane change.

**Lemma 1** *(Collision-Free Dynamics) If the safety distance threshold $d_{\min}$ is strictly enforced for all lane changes, then any sequence of lane-change actions will be collision-free. Formally, for every state $s \in \mathcal{S}$: if the ego vehicle intends to change into a target lane $l'$, and for all vehicles $i$ in the relevant positions $J_{l'}$ of lane $l'$, presence of a vehicle $i$ implies a distance of at least $d_{\min}$ (i.e. $\forall i \in J_{l'} :; \text{presence}_i \Rightarrow \text{distance}_i \geq d_{\min}$), then executing that lane change does not cause a collision.*

***Proof*** We prove this by induction on the number of lane changes. If no lane changes are made, collisions cannot occur. Assume as an induction hypothesis that any sequence of $k-1$ lane changes respecting the distance condition is collision-free. Now, consider a sequence of $k$ lane changes under the same rule. Let the first $k-1$ changes produce no collision (by the hypothesis). For the $k$-th lane change, the precondition guarantees that every nearby vehicle in the target lane $l'$ is at least $d_{\min}$ away from the ego vehicle prior to merging. By Definition 12 and the distance condition, there will be no spatial overlap between the ego vehicle and any vehicle in lane $l'$ during the merge. This means the $k$-th lane change itself is collision-free. Consequently, all $k$ lane changes have been executed without any collision. □

**Lemma 2** *(Data Collection Validity) The ego vehicle must ensure that data collection from surrounding vehicles is complete for the decision-making algorithm to be executable:*

$$Data\ \ Collection = \bigwedge_{i=1}^{n} datacheck_i > 0$$

*where $datacheck_i$ indicates successful data acquisition (e.g., presence and speed) of the surrounding vehicles $i$. The lane-change decision algorithm is enabled only in states where Data Collection holds true.*

***Proof*** The decision-making algorithm takes as input the state of surrounding vehicles (such as whether each position is occupied and the speeds of those vehicles). If any required datum $\text{datacheck}_i$ is missing, then the algorithm's state is incomplete. In that case, critical variables in the decision computations (such as safety gaps or incentive values) are undefined or unreliable. Consequently, no valid decision can be determined by the algorithm. In formal terms, if $\neg$Data Collection, then the transition function $\mathcal{P}(s, a)$ for the ego vehicle's decision $a$ is not enabled (or leads to an undefined outcome). This effectively causes a deadlock or a pause in decision-making until the data is complete. Therefore, a necessary precondition for the algorithm to operate is Data Collection being true. The ego vehicle's controller must ensure all $n$ data checks are positive before making any lane-change decision. In summary, complete data

collection is required for the algorithm to proceed, which justifies the given predicate. □

**Proposition 1** (Optimization Objective) Let $\mathcal{M} = (\mathcal{S}, s', \mathcal{A}, \mathcal{P}, AP, L)$ be an MDP as in Definition 1, and let $\mathcal{R} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}_{\geq 0}$ be a reward structure for $\mathcal{M}$ as in Definition 3. We assume there is a "safety" condition $\varphi_{\text{safe}} \subseteq (2^{AP})^{\omega}$ that must hold at all times (an invariant). Then the *lane-changing algorithm* aims to find a policy $\pi$ maximizing:

$$J(\pi) \;=\; E_{\pi}\Big[\sum_{t=0}^{T} \mathcal{R}(s_t, a_t, s_{t+1})\Big],$$

subject to the constraint that the safety conditions hold at all times. In other words, find $\pi^* = \arg\max_{\pi} J(\pi)$ such that $\Pr_{\pi}(\text{always } \varphi_{\text{safe}}) = 1$.

***Proof*** We reduce the lane-changing problem to an MDP optimization problem with constraints. By Definition 1, an MDP captures all possible state transitions under probabilistic and nondeterministic choices. The ego's choices of maneuvers (actions) correspond to a strategy $\pi$ resolving nondeterminism (Definition 2). Under any fixed strategy $\pi$, the induced behavior is a Markov chain, and $J(\pi)$ is the expected total reward accrued over $T$ steps (a finite horizon). Here $E_{\pi}[\cdot]$ is the expectation w.r.t. the probability distribution on paths determined by $\pi$ (for formality, if $\Sigma$ is the set of paths, $E\pi[X] = \int_{\Sigma} X(\pi), d\Pr_{\pi}(\sigma)$). The safety constraints (such as maintaining safe distances and collision avoidance) are encoded as a state predicate $\varphi_{\text{safe}}$ (as used in the PCTL formula $P_{\geq 1}[G, \varphi_{\text{safe}}]$). Requiring safety means we restrict attention to safe policies – those that never lead to a state violating $\varphi_{\text{safe}}$. In practice, this can be ensured by disallowing unsafe actions (or equivalently assigning them a penalty in $\mathcal{R}$ so they are never optimal).

Under these conditions, the optimization problem becomes: choose a strategy $\pi$ (from the set of all policies that satisfy the safety invariant) that maximizes $J(\pi)$. Standard results in MDP theory guarantee that an optimal strategy $\pi^*$ exists for finite-horizon MDPs with finite state/action spaces. For example, one can apply dynamic programming or linear programming to compute the highest expected reward, and because the rewards are bounded and the policy space can be restricted to measurable (even deterministic Markov) policies, an optimum is attained. Intuitively, since $T$ is finite, we can perform backward induction on the MDP's state space to compute the best possible cumulative reward from each state while never choosing an unsafe action. This yields $\pi^*$. By construction, $\pi^*$ maximizes expected reward while never violating safety constraints. Thus, we have formally established that the lane-changing control problem can be posed as finding $\pi^*$ that maximizes $E_{\pi}\left[\sum_{t=0}^{T} \mathcal{R}(s_t, a_t, s_{t+1})\right]$ subject to the invariant $\varphi_{\text{safe}}$. This aligns with the stated objective: achieve the highest possible cumulative reward without ever compromising safety. □

The notions above provide the main components for our lane-changing setting. We now state the closed-loop semantics that connect the controller, the longitudinal dynamics, and the environment MDP in a way that PRISM can analyze directly.

Let $X$ be the set of traffic states observed by the ego vehicle. Let $A = \{\text{keep}, \text{left}, \text{right}\}$ be the action set. Let $f_{\text{long}}$ be a longitudinal dynamics model that maps local states to accelerations for the ego vehicle and its neighbors. Examples of $f_{\text{long}}$ include IDM, Gipps, and calibrated car-following models.

A lane-change strategy $S$ is a decision rule that maps an observation $o(x)$ with $x \in X$ to a set of admissible actions $D(x) \subseteq A$. Actions are admissible when two predicates are satisfied

$$\text{incentives}_S(x;\theta) = \text{true}, \qquad \text{safety}_S(x;\theta) = \text{true},$$

where $\theta$ are the parameters of the strategy. The predicates are expressed using the accelerations produced by $f_{\text{long}}$ in state $x$.

We compile $S$ into guards

$$\begin{aligned}
&\text{can_left}(x) \equiv \text{target_left_free}(x) \;\wedge\; \text{incentives}_S(x;\theta) \;\wedge\; \text{safety}_S(x;\theta),\\
&\text{can_right}(x) \equiv \text{target_right_free}(x) \;\wedge\; \text{incentives}_S(x;\theta) \;\wedge\; \text{safety}_S(x;\theta),\\
&\text{can_keep}(x) \equiv \neg\,\text{can_left}(x) \;\wedge\; \neg\,\text{can_right}(x) \;\text{ or a safe-following condition.}
\end{aligned}$$

The environment MDP $M = (X, A, P)$ describes stochastic traffic evolution and sensing. The closed-loop model $M^S$ is obtained by restricting $M$ to the actions enabled by these guards, with a fair randomized tie break when both lateral moves are admissible. In the next section, the formulation of MOBIL is an instance of $S$ that provides concrete incentive and safety predicates, and IDM is an instance of $f_{\text{long}}$ that supplies accelerations. The distinction is explicit. MOBIL is the algorithm, IDM is part of the dynamics model, and the model checker encodes the closed-loop interaction model $M^S$.

## 5.2 Verification Properties for Lane-Changing Algorithms

Next, we propose and formalize key properties necessary for verifying lane changing algorithms, providing a

**Table 1** Verification properties for lane-changing algorithms

| Properties | Specifications | Specifications in PCTL |
|---|---|---|
| Safety Properties | Freedom of Deadlock | $\exists\,[\neg G(deadlock)]$ |
| | Satisfaction of Safety Criterion | $\mathrm{P}_{\geq 1}[\mathrm{G}(\phi_{safe})]$ |
| Liveness Properties | Eventual Progress | $\mathrm{P}_{>0}[\mathrm{F}(action \in stay, Change\ Lane)]$ |
| | Avoiding Stagnation | $\mathrm{P}_{>0}[\mathrm{F}(\phi_{safe})]$ |
| | Lane Change Feasibility | $\mathrm{P}_{>0}[\mathrm{F}(action = change \wedge \phi_{safe_{target}})]$ |
| Performance Properties | Time Efficiency | $E_{max}[\text{“}Time\ Efficiency'']$ |
| | Minimizing Critical State | $E_{min}[\text{“}Time\ In\ Critical\ States'']$ |
| | Avoiding Excessive Deceleration | $E_{min}[\text{“}Total\ Deceleration'']$ |

rigorous basis for proving their correctness and robustness in dynamic environments.

### 5.3 Instantiating Lane–change Strategies

The verification properties in Table 1 form a common suite that we apply to any concrete controller. We now explain how a strategy is compiled into the closed–loop model so that the same properties and reward structures can be evaluated without additional modeling effort.

We abstract lane–change logic as two predicates over accelerations

$$\mathrm{incentives}_S(x;\theta) \qquad \text{and} \qquad \mathrm{safety}_S(x;\theta),$$

where $x$ is the local traffic state and $\theta$ are the strategy parameters. The compilation–as–guards procedure turns any strategy $S$ into a formal model that fits the closed–loop semantics described above.

The procedure of CaaG procedure follows:

1. Choose a longitudinal model $f_{\mathrm{long}}$ such as IDM, Gipps, Cooperative Adaptive Cruise Control (CACC), or a learned policy. This choice fixes how accelerations for the ego vehicle and its neighbors are computed.
2. Express the incentive and safety predicates of $S$ using the accelerations produced by $f_{\mathrm{long}}$. Examples include:

- ➢ *MOBIL:* An incentive inequality of the form $\Delta a_{\mathrm{ego}} + p(\Delta a_{\mathrm{newf}} + \Delta a_{\mathrm{oldf}}) > a_{\mathrm{thr}}$ together with a follower–safety bound $a_{\mathrm{follower}} \geq -b_{\mathrm{safe}}$.
- ➢ *Gap–acceptance:* Spatial rules such as $\mathrm{gap} > \mathrm{min_gap}$ and $\hat{\mathrm{gap}} > \mathrm{min_closure}$ are translated into acceleration inequalities via $f_{\mathrm{long}}$.
- ➢ *Risk minimization:* Choices of the form $\arg\ min_{d\in D}\mathbb{E}[\mathrm{risk}(x,d)]$ are compiled as guard thresholds or as reward–optimal selections.

3. Introduce transitions labeled `left`, `right`, and `keep`. Enable each transition with the corresponding guards $\mathrm{can_left}$, $\mathrm{can_right}$, and $\mathrm{can_keep}$. Resolve joint admissibility with a fair randomizer to avoid scheduler artifacts.
4. Reuse the same PCTL queries and reward specifications defined earlier. This yields safety reachability, justified lane changes, density–conditioned rate, efficiency, and comfort, all under min–max semantics.

Only the rule–lifting step depends on the chosen strategy, while other steps are reusable. In Section 6 we instantiate MOBIL by supplying its incentive and safety predicates and by selecting IDM as the longitudinal plug in. This demonstrates that the framework is algorithm independent and that it enables direct and reproducible comparison across controllers within the same closed loop semantics.

## 6 Case Study: Formalization and Verification of MOBIL

This section provides the formal foundations for analyzing and verifying the MOBIL algorithm. It introduces key definitions, lemmas, and propositions based on the foundations from the previous section to formalize the algorithm's properties, including safety, feasibility, and performance, ensuring effective operation in dynamic traffic scenarios.

### 6.1 Formalization of MOBIL

The MOBIL model [13] is being considered as a de facto benchmark lane-changing model in traffic simulations and, as such, is a useful proxy for understanding and evaluating automated driving behaviors. Therefore, analysis results from MOBIL can be viewed as an informative starting point for the design and verification of more advanced

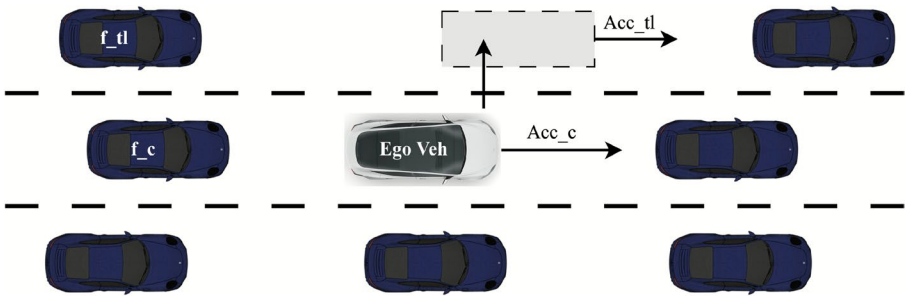



**Fig. 2** Ego vehicle considering lane change to left as vehicle is in front (old and new followers are denoted by $f_c$ and $f_{tl}$ respectively).

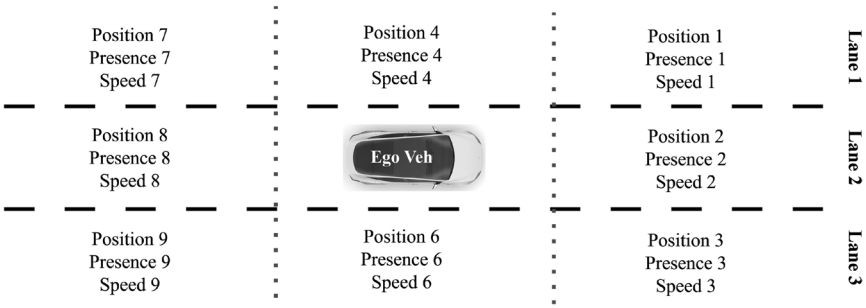



**Fig. 3** Relative positions of surrounding vehicles in the discretized $3 \times 3$ neighborhood around the ego vehicle. The ego occupies the center cell and is not counted among the indexed neighbors. Cells are indexed 1..9 except 5, as shown, spanning left, current, right lanes crossed with ahead, alongside, behind.

lane-changing algorithms toward safer and more reliable autonomous driving systems. The MOBIL is formalized where the speed of the ego vehicle is discretized with three distinct states as defined in Definition 7. This abstraction makes the continuous nature of speed manageable for our lane-changing decisions by simplifying it into manageable categories. It discretizes the dynamics and simplifies its analysis while preserving the important dynamics to the model to capture all the corner cases.

**Definition 13** (Ego Vehicle State in MOBIL) An ego vehicle in the MOBIL model is characterized by a state tuple $(s, l, v, a)$, where: $s \in \{0, 1, 2, 3, 4, 5, 6, 7\}$ represents the operational state, $l \in \{1, 2, 3\}$ denotes the current lane number, $v \in \{1, 2, 3\}$ represents the discretized speed level: $v = 1 : 0 - 40km/h$,$v = 2 : 41 - 80km/h$, and $v = 3 : 81 - 120km/h$ and $a \in \{0, 1\}$ indicates whether the ego vehicle has collected environmental data $(a = 1)$.

Building upon Definition 8 of surrounding vehicles, we define specific positions relative to the ego vehicle that are critical for lane-changing decisions in MOBIL. We have made some assumptions that an ego vehicle only considers surrounding vehicles within its unsafe distance. Relative positions from vehicles across three lanes are identified to locate surrounding vehicles.

For example, the surrounding vehicles of an ego vehicle in the middle lane (Lane 2) are at positions 1,2,3,4,6,7,8, and 9, depending on their presence. The ego vehicle position in the center lane is depicted in Fig. 2, and the corresponding position of the surrounding vehicles is also shown in Fig. 3. Consider the case where the vehicle that is directly in front of the ego vehicle (Position 2) is within an unsafe distance. In this scenario, the ego vehicle has to do a lane change.

In our formalization, both the speed and the relative positions are discretized to simplify the decision-making process. The speed of the ego vehicle is divided into three levels, while the speeds of the surrounding vehicles are classified on a scale from 0 to 4, with 0 indicating the absence of a vehicle. Furthermore, the relative positions of the surrounding vehicles are fixed within a $3 \times 3$ grid around the ego vehicle, represented by indices 1 to 9. Thus, the qualitative situation shown in Fig. 2 becomes a Boolean presence map, along with discrete speed values in the indexed cells of Fig. 3 (e.g., Position 2 = "ahead in current lane", Position 9 = "behind in right lane"). This is consistent with our formal definitions and the PRISM labels used in the formalizations.

This detailed enumeration of positions ensures MOBIL comprehensively accounts for all relevant surrounding vehicles during decision-making.

**Definition 14** (Surrounding Vehicles in MOBIL) For each relevant position relative to the ego vehicle, MOBIL defines: $presence_i \in \{true, false\}$ indicates the presence of a vehicle at position $i$ and $speed_i \in \{0, 1, 2, 3, 4\}$ is discretized speed of the vehicle at position $i$, where $speed_i = 0$ is when no vehicle is present, or data not collected, and $speed_i = 4$ is $121 - 160km/h$. Other speeds are the same as for ego vehicle defined in Definition 13.

In the above definition of surrounding vehicles, the $speed_i = 4$ corresponds to velocities in the range $121 - 160km/h$. This upper range is included to capture cases in which other vehicles may exceed typical operating

speeds or violate posted limits, thereby ensuring the model can encompass the full spectrum of observed traffic behavior in real-world scenarios.

**Definition 15** (Politeness Factor) The Politeness Factor $p$ in MOBIL reflects the ego vehicle's consideration for the acceleration changes of other drivers. $p = 0$: Egoistic behavior, where only ego vehicle's benefits are considered in lane-changing decisions. $p = 1$: Altruistic behavior, where the ego vehicle also accounts for the impact of its action on surrounding vehicles' acceleration.

**Definition 16** (Incentive Criterion in MOBIL) The incentive criterion in MOBIL evaluates the net benefit of lane change by considering both the ego vehicle's acceleration and the impact on surrounding vehicles:

$$\textit{Incentive Criterion} = \Delta acc_{ego} + p(\Delta acc_{new_follower} + \Delta acc_{old_follower}) > a_{thr}$$

here, p is the politeness factor, which is defined in Definition 15.

**Definition 17** (Safety Criterion in MOBIL) The safety criterion in MOBIL ensures that any lane change does not adversely affect the safety of surrounding vehicles:

$$Safety\ Criterion = acc_{ft} \geq b_{safe}$$

More specifically, for the left and right lane,

$$Safety_left = acc_{ft_l} \geq b_{safe} \text{ and } Safety_right = acc_{ft_r} \geq b_{safe}$$

Where $acc_{ft}$is the acceleration of new follower in target lane (left or right) after the lane change and $b_{safe}$ is the safety threshold (e.g, $b_{safe} = -1$) representing the maximum acceptable deceleration to prevent unsafe conditions. We use the IDM to compute longitudinal accelerations. The IDM acceleration is

$$a_{\mathrm{IDM}}(v, \Delta v, s) \;=\; a_0 \left[ 1 - \left( \frac{v}{v_0} \right)^{\delta} - \left( \frac{s^*(v, \Delta v)}{s} \right)^2 \right],$$

$$s^*(v, \Delta v) \;=\; s_0 + vT + \frac{v\,\Delta v}{2\sqrt{a_0\, b}},$$

where $v$ is ego speed, $\Delta v$ is ego minus leader speed, $s$ is the gap, $v_0$ is desired speed, $a_0$ is maximum acceleration, $b$ is comfortable braking, $s_0$ is the jam distance, $T$ is the time headway, and $\delta$ is typically 4. We discretize $v$, $\Delta v$, and $s$ to the model's levels and map $a_{\mathrm{IDM}}$ to the acceleration categories used in the guards.

The incentive and safety predicates use the accelerations from the dynamics, as specified in Definitions 16 and 17, respectively. All accelerations on the right-hand side are computed with the IDM on the current and post-change configurations. This provides a clear distinction: the lane-change rule is the algorithm, IDM is the dynamics model, and the closed-loop MDP $M^S$ is what the proposed approach allows to verify.

**Proposition 2** (Lane Changing Decision Rule in MOBIL) Let the Incentive Criterion and Safety Criterion be those of Definitions 16 and 17, respectively. Under the MOBIL model, an ego vehicle initiates a lane change to lane if and only if all three of the following conditions hold:

1. The Incentive Criterion exceeds the threshold $a_{thr}$.
2. The Safety Criterion is satisfied.
3. The target lane $l'$ is available.

***Proof*** ($\Rightarrow$) If all conditions are satisfied, then a lane change is initiated. Assume conditions (1), (2), and (3) are all true for a potential move to lane $l'$. Condition (1) guarantees a net benefit for changing lanes (e.g., improved speed or traffic flow), condition (2) guarantees the maneuver can be done safely, and condition (3) ensures there is a feasible lane to move into. In this case, all requirements for a lane change are met, so the MOBIL decision rule will permit and initiate the lane change to $l'$. In other words, the guard for the lane-change action (in a formal model) evaluates to true, enabling the ego vehicle to execute the lane change.

($\Leftarrow$) If a lane change is initiated, then all conditions must have been satisfied. Now, assume the ego vehicle does initiate a lane change to $l'$. We argue that none of the conditions (1)–(3) could have been false in this scenario. If the Incentive Criterion were not above $a_{thr}$ (condition (1) false), the maneuver would offer insufficient benefit or even be unsafe, contradicting the assumption that the algorithm would rationally choose to change lanes. If the Safety Criterion did not hold (condition (2) false), proceeding with the lane change would create an unsafe situation (e.g., forcing a following vehicle to brake dangerously or causing a collision), which the algorithm is designed to avoid. If the lane were not available (condition (3) false), then the lane change would be physically impossible or illegal (for example, there might be no lane $l'$ to the side, or it might be blocked), so the ego vehicle could not have completed the maneuver. In any of these cases, initiating the lane change would violate the algorithm's rules or physical constraints. Thus, a lane change occurring implies that no feasibility condition was violated – i.e., all three conditions must indeed have been true. □

**Lemma 3** *(Safety Precedence) If the Safety Criterion is not satisfied, the ego vehicle must not change lanes – regardless of the value of the Incentive Criterion.*

***Proof*** Suppose the Safety Criterion is violated for a potential lane change. In this situation, a lane change is forbidden, no matter how large the incentive might be. Formally, when the safety condition is false, one of the feasibility requirements for a lane change is missing. By the decision rule (proved above), a lane change cannot be initiated if any required condition fails – in particular, a false Safety Criterion in Definition 17 prevents the maneuver. Therefore, the ego vehicle must refrain from changing lanes if safety is not ensured. Intuitively, even if a lane change promises a gain (higher speed or other benefit), that gain is overridden by the risk of collision or forcing other vehicles into dangerous braking. Safety takes precedence over incentive: the algorithm will simply not allow the lane change in unsafe conditions. In our PRISM formalization of MOBIL, this principle is built in by guarding all lane-change actions with the Safety Criterion. Thus, whenever the Safety Criterion is false, the transition for changing lanes is disabled, reflecting that safety considerations override any incentive to change lanes. □

**Lemma 4** *(Collision-Free Lane Changes) If the safety criterion is strictly enforced, then lane changes will always be collision-free:*

$$\forall s \in \mathcal{S}, Safety_{left} \vee Safety_{right} \Rightarrow no\ \ Collisions$$

***Proof*** The Safety Criterion ensures that a lane change is executed only when it is safe. Suppose, for contradiction, that a collision occurs despite strictly enforcing this criterion. For a crash to happen, at least one surrounding vehicle must have been too close or forced to decelerate beyond the safe threshold $b_{safe}$, contradicting the criterion's enforcement. Since every lane change must satisfy the Safety Criterion, each maneuver remains collision-free, preserving a safe traffic state. By induction, no lane change under strict enforcement ever results in a collision. The PRISM model confirms this: paths respecting safety constraints never reach a collision state, verifying that safety-enforced lane changes are always collision-free. □

**Definition 18** (Critical State) A Critical State in MOBIL occurs when the ego vehicle cannot safely continue in the current lane or execute a safe lane change. Formally:

$$Critical\ State\ =\ \neg Safety_{Current} \wedge \neg(Safety_{left} \vee Safety_{right})$$

In such a state, the vehicle may need to perform emergency maneuvers or take corrective actions to maintain safety.

**Lemma 5** *(Effect of Traffic Density on Politeness factor) Increasing the politeness factor p in MOBIL generally leads to fewer lane changes by the ego vehicle, although the relationship can be influenced by traffic density and extreme parameter values. Specifically, as p rises, the ego vehicle becomes more considerate of other drivers' acceleration changes, which usually reduces its willingness to change lanes for its own benefit. However, under certain conditions (e.g., very high p with low traffic or modeling discretizations), counter-intuitive effects can occur.*

***Proof*** As established in Definition 16, the Incentive Criterion in MOBIL requires $\Delta a_{ego} + p \cdot (\Delta a_{new_follower} + \Delta a_{old_follower}) > a_{\text{thr}}$. Here $p \in [0, 1]$ (Definition 15) interpolates between purely egoistic behavior ($p = 0$ considers only the ego vehicle's acceleration gain $\Delta a_{ego}$) and fully altruistic behavior ($p = 1$ gives full weight to the total acceleration impact on others). Typically, lane changes cause a net deceleration for surrounding vehicles ($\Delta a_{others} < 0$), leading to a decreasing function $f(p) = \Delta a_{ego} + p \cdot \Delta a_{others}$. As $p$ increases, $f(p)$ decreases, making the incentive criterion harder to satisfy and reducing lane-change frequency. This aligns with empirical findings where higher politeness leads to fewer lane changes due to increased hesitancy in disrupting traffic.

However, corner cases arise. When $\Delta a_{others} \geq 0$ (e.g., the ego vehicle's move benefits its old follower or when the target lane is empty), $f(p)$ increases with $p$, potentially encouraging lane changes. At extreme $p$, counterintuitive effects emerge: a very polite vehicle might change lanes in scenarios where a selfish one would not, or an overly cautious ego may hesitate excessively, causing late, abrupt maneuvers. Additionally, discrete speed modeling can create non-monotonic lane-changing patterns. In summary, while increasing $p$ generally reduces lane changes, the effect is modulated by traffic density and specific model conditions, confirming the lemma's refined claim. □

## 6.2 Verification Results

In this section, we assess the correctness and performance of MOBIL based on the proposed probabilistic model checking approach. We mainly verify the safety, liveness and performance properties under different MOBIL parameter configurations by building on the generic properties outlined in Section 6.1.

#### 6.2.1 Safety Properties

We have verified some safety properties to ensure the safety of the ego vehicle, which are:

**Deadlock Freedom** A deadlock freedom property ensures that ego vehicle will not indefinitely wait to change lanes due to conflicting conditions or resource availability. Hence, we verify the following property to ensure that.

$$\exists\, [G\ \neg\text{“deadlock”}]$$

In PRISM, we defined this as E [G ¬“deadlock″], where E is ‘there exists’, G is ‘globally’, and deadlock is an atomic proposition that comes built in within PRISM.

**Safety Threshold and Critical States** We analyse the minimum reachability probability of a critical configuration by verifying $P_{\min} =?\,[\,F\ \text{"Critical_State"}\,]$ while sweeping the follower-braking threshold $b_{\text{safe}}$ used in the safety guard. For $b_{\text{safe}} = -2$ and $-3$ or greater, the model checker returns $P_{\min} = 1.0$, which implies that an adversarial scheduler can force the system into a critical state from the initial condition. In contrast, for $b_{\text{safe}} = -4$ and $-5$, the result is $P_{\min} = 0.0$, which means that the critical set becomes unreachable under all schedulers. Taken together, these observations identify a safety transition at $b_{\text{safe}} = -4$: once the guard permits comfortable braking consistent with IDM, the abstraction prevents any adversarial strategy from reaching a critical state. The individual verification outcomes are shown in Fig. 4a-c.

**Recovery from Unsafe Conditions** For $b_{\text{safe}} > -4$, we verify the bounded-recovery requirement

$$P \geq 0.90\big[\,\text{"Critical_State"}\ \mathrm{U}^{\leq 10}\ (s = 0)\,\big].$$

From states labelled `"Critical_State"`, this asserts that the ego returns to the nominal mode $s = 0$ within at most ten steps with probability at least 0.90, while remaining in the critical region until recovery. With the label

$$\text{label "Critical_State"} \equiv (s = 7),$$

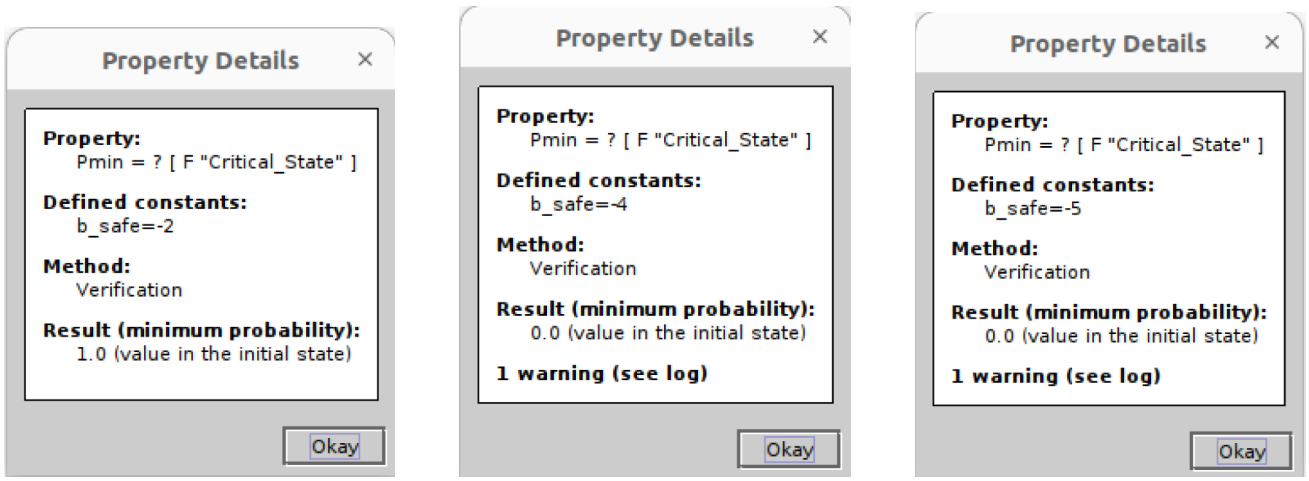
PRISM verifies the property as true for $b_{\text{safe}} > -4$. The result indicates that the controller does not remain indefinitely in an unsafe configuration and promptly regains safety with high probability.

#### 6.2.2 Liveness Properties

Here, we present some liveness properties that ensure that the ego vehicle is doing progress. Eventual Progress

$$P \geq 1\,[\mathrm{F}\ (\text{“Lane_Changed”} \vee \text{“Keeping_Lane”})]$$

This property guarantees with probability 1 that the system will ultimately change lanes or remain in its lane.“Keeping_Lane ” and “Lane_Changed” are the representation of the states in




(a) $b_{\text{safe}} = -2$: $P_{\min} = 1.0$ (b) $b_{\text{safe}} = -4$: $P_{\min} = 0.0$ (c) $b_{\text{safe}} = -5$: $P_{\min} = 0.0$

**Fig. 4** Minimum probability of eventually entering a critical state for different follower-braking thresholds $b_{\text{safe}}$.

which the vehicle has been keeping the lane and changing the lane, respectively, in the model and given as:

label "Lane_Changed" $= (s = 2) \vee (s = 3)$;

label "Keeping_Lane" $= (s = 1)$;

In the PRISM model checker, this property holds "true", ensuring that the system progresses to one of these two outcomes.

Eventually Make a Justified Lane

$$\mathcal{G}_{max} =? [\mathrm{F}\ ((incentive_left \vee incentive_right) \wedge\ \text{“}Lane_Changed\text{”})]$$

This property computes the maximum probability that, at some later time in the future, the system performs a lane change action, i.e., (Lane_Changed) with incentive (incentive_left or incentive_right) to change the lane. This makes sure that lane change takes place only with the existence of a valid incentive. The incentive formulas are defined as:

$$\text{formula incentive_left} = \Delta\text{acc}_{\text{ego_left}} + p \cdot (\Delta\text{acc}_{\text{new_follower_left}} + \Delta\text{acc}_{\text{old_follower}}) > a_{thr}$$

$$\text{formula incentive_right} = \Delta\text{acc}_{\text{ego_right}} + p \cdot (\Delta\text{acc}_{\text{new_follower_right}} + \Delta\text{acc}_{\text{old_follower}}) > a_{thr}$$

PRISM returns $\mathsf{P}_{\mathsf{max}} = 1$, indicating that the ego vehicle is guaranteed to eventually perform a justified lane.

### 6.2.3 Performance Properties

Using reachability rewards [29] in PRISM, we evaluate our lane-changing model (MOBIL) under different parameter configurations for various reward-based properties. Two of these properties, in particular, will guide our selection of the optimal parameter values.

We leverage upon cumulative reward properties to find the minimum expected number of lane changes by ego vehicle within time bound T in our model.

$$R\{Lane_Changes\}_{min} =? [\mathsf{C} \leq \mathsf{T}]$$

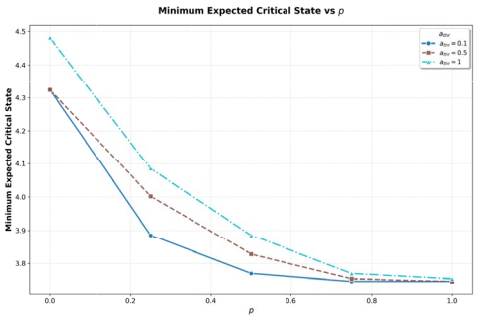
Similarly,

$$R\{Critical_State_Penalty\}_{min} =? [\mathsf{C} \leq \mathsf{T}]$$

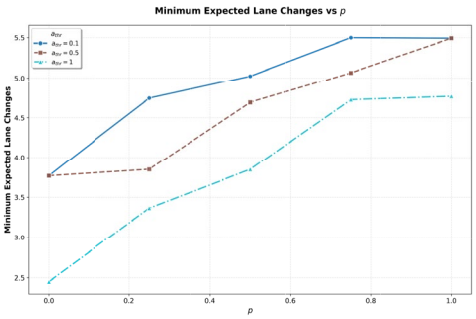
which measures the minimum expected critical state penalty accumulated before termination within time bound T. In the properties specified above in PCTL, $R$ denotes the expected cumulative reward.

We instantiated the model for $T = 200$ steps and swept the MOBIL politeness factor $p \in \{0, 0.25, 0.50, 0.75, 1\}$ and the acceleration threshold $a_{\text{thr}} \in \{0.1, 0.5, 1\}$. Figure 5a reports the minimum expected critical-state penalty, and Fig. 5b reports the minimum expected number of lane changes.

Across all $a_{\text{thr}}$, increasing $p$ reduces the expected critical-state penalty and the curves approach a common value near 3.75 as $p \to 1$ (Fig. 5a). In contrast, the minimum expected lane changes increase with $p$, with lower $a_{\text{thr}}$ yielding more maneuvers for any fixed $p$ (Fig. 5b). The fewest maneuvers occur at $(p = 0,\ a_{\text{thr}} = 1)$ but this setting exhibits the




(a) Minimum expected critical-state penalty as a function of politeness $p$ for three incentive thresholds $a_{\text{thr}}$. Larger $p$ consistently lowers the penalty, and the three curves converge for $p \geq 0.75$.




(b) Minimum expected lane changes as a function of politeness $p$ for three incentive thresholds $a_{\text{thr}}$. Maneuvers increase with $p$, and lower $a_{\text{thr}}$ produces more lane changes for a fixed $p$.

**Fig. 5** Comparison of minimum expected critical-state penalty and minimum expected lane changes as a function of politeness $p$ for three incentive thresholds $a_{\text{thr}}$.

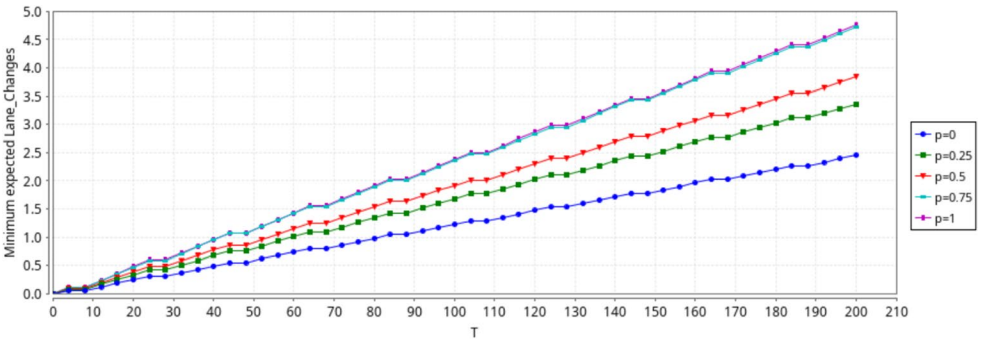



**Fig. 6** Expected lane changes over time steps.

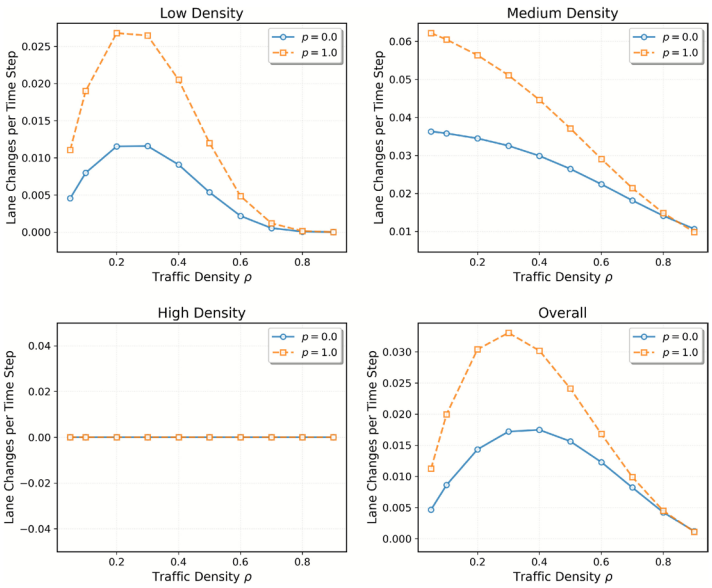



**Fig. 7** Lane-change frequency across traffic densities under varying politeness parameters.

highest penalty. The lowest penalties are achieved at higher $p$ or at lower $a_{\text{thr}}$, at the cost of additional lane changes.

**Minimum Expected Number of Lane Changes** One key metric examined is the Minimum Expected Number of Lane Changes over 200 steps.

$$R\{Lane_Changes\}_{min} =? \; [\mathsf{C} \leq 200]$$

Contrary to intuition, a highly polite AV $(p = 1)$ executes more lane changes, as it carefully evaluates surrounding traffic and waits for opportunities that benefit both itself and others. In contrast, an egoistic AV $(p = 0)$ remains in its lane longer, making fewer but riskier lane changes.

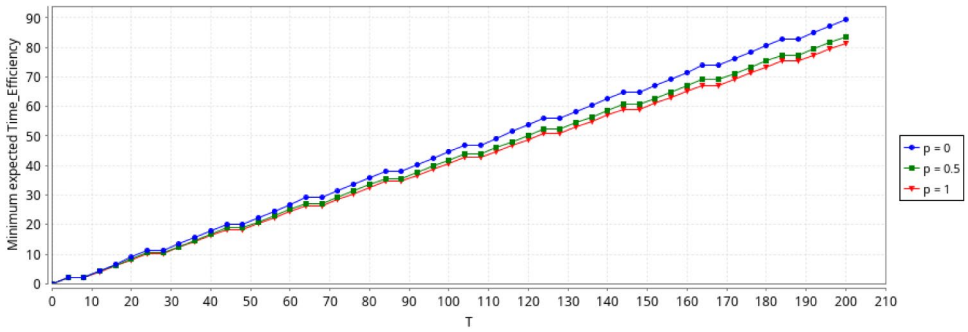
This result is not erroneous but reflects the mathematical rigor of the model: higher politeness $(p)$ promotes lane changes only when they are beneficial for both the ego vehicle and surrounding traffic. The embedded safety constraints eliminate hazardous maneuvers, ensuring that only safe and beneficial lane changes occur. Thus, selflessness $(p = 1)$ fosters cooperative driving, leading to frequent, yet safer and more efficient lane changes (Fig. 6). However, the optimal balance is found at moderate politeness $(p = 0.5)$, ensuring safety without excessive lane-switching. These findings validate Lemma 5, demonstrating the influence of traffic density on the politeness factor.

**Lane Change Rate versus Traffic Density** We report lane–change *rates* via density–conditioned reward ratios

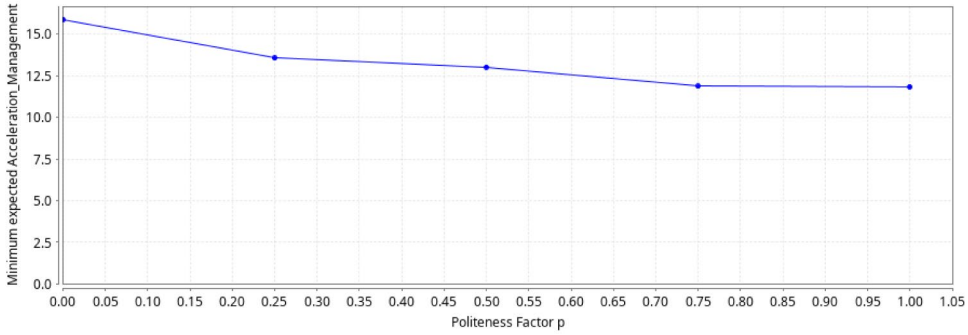
$$r_{\text{bin}} = \frac{R\{\text{"LC_bin"}\}_{\max} =? \, [C \leq T]}{R\{\text{"Time_bin"}\}_{\max} =? \, [C \leq T]},$$

which normalizes the count of lane changes relative to the time spent in each density bin. Here, LC_bin denotes the number of lane changes across different traffic density levels (e.g., low, medium, high), while Time_bin represents the duration of time spent within each density bin.

The set of four plots in Fig. 7 demonstrates the canonical distribution of lane-changing activity across low, medium, high, and aggregated density regimes, characterized by a pronounced peak at intermediate densities and a subsequent decline under saturated traffic conditions. This behavior reflects the well-established principle that lane-changing maneuvers are most prevalent when traffic gaps frequently emerge and dissipate, but




(a) Time efficiency at different $p$ values.




(b) Minimum deceleration over different politeness factors.

**Fig. 8** Comparison of time efficiency and minimum deceleration as functions of politeness $p$.

diminish as roadway capacity becomes saturated and viable opportunities vanish. In the low- and medium-density regimes, an increase in the politeness factor $p$ amplifies the frequency of lane changes, as follower-benefiting maneuvers more consistently fulfill both incentive and safety constraints. Conversely, in the high-density regime, all trajectories converge toward zero, signifying a scarcity of permissible gaps. These observed dynamics are consistent with MOBIL-based investigations, which similarly identified a marked mid-density maximum and strong dependence on politeness sensitivity [13]. Importantly, our adoption of reward-ratio semantics mitigates sampling-related artifacts, thereby facilitating rigorous cross-strategy comparisons and enhancing the interpretability of model instantiations.

**Time Efficiency and Acceleration Management** We evaluate system performance through the lenses of time efficiency and driving comfort, certified against worst-case nondeterminism. We compute the minimum expected time efficiency, $R\{\text{"Time_Efficiency"}\}_{\min} =?[C \leq T]$, to establish a guaranteed lower bound on traffic flow. Furthermore, we evaluate the minimum expected acceleration management, $R\{\texttt{"Acceleration_Management"}\}_{\min} =?[F($ `"Lane_Changed" V"Critical_State")`$]$, to quantify the smoothness of driving maneuvers.

The results, shown in Fig. 8a and b, reveal a clear trade-off governed by the politeness factor $p$. Time efficiency shows a mild decrease at larger $p$ (Fig. 8a), consistent with longer cooperative waits before a safe merge is deemed acceptable. Conversely, the acceleration-management metric improves significantly at moderate $p$ (Fig. 8b), reflecting smoother decelerations and fewer strong-brake events as cooperation increases. This identifies an optimal balance at a politeness factor of $p = 0.5$. While $p = 0$ leads to aggressive, inefficient behavior, and $p = 1$ reduces deceleration at the cost of both safety and efficiency, $p = 0.5$ ensures stable and smooth driving. Unlike open-loop simulation, these values are certified as min-optimal under all schedulers, meaning both efficiency and comfort guarantees hold against worst-case nondeterminism. These findings emphasize the importance of tuning cooperation to achieve a practical and robust trade-off for autonomous driving.

Based on the comprehensive verification results presented, we conclude that the proposed probabilistic model checking framework successfully captures the nuanced behavior of the MOBIL lane-changing algorithm and

provides rigorous, scheduler-robust guarantees that complement and enhance findings from traditional simulation-based studies.

Lane-changing models or algorithms, such as MOBIL, have been extensively analyzed both analytically and empirically over the past decade. Early work, including Kesting et al. [13], established that lane-change frequency exhibits non-monotonic dependence on traffic density, peaking at moderate flow (approximately 10 to 15 vehicles per kilometer per lane) and declining in high congestion. Subsequent studies, for example Khan et al. [30], extended these analyses, demonstrating that increased politeness and stricter acceleration thresholds reduce the frequency of lane changes and improve overall safety. More recent developments continue to support this trend. Abtahi and Azim [31] introduced a "Minimizing Lane Changes" state-machine algorithm, which simulation results indicated achieved a 33–50% reduction in lane-change frequency compared to MOBIL, along with fewer collisions. This underscores the safety versus efficiency trade-off associated with MOBIL's more frequent lane changes. Similarly, Wang et al. [32] demonstrated that a deep reinforcement learning-based decision policy can outperform rule-based models by decreasing collision rates and delivering smoother, more stable lane-changing behavior while maintaining traffic efficiency. Collectively, these findings reinforce the conclusion that minimizing unnecessary lane changes enhances safety and contributes to overall traffic stability.

The formal analysis presented in this work confirms several well-established qualitative trends from this body of literature. The density-conditioned rewards produce lane change rate versus traffic density curves that reproduce all first-order features found in MOBIL micro-simulation [13]. The rates rise from light traffic levels to peak at an intermediate density band, then decay as density increases further. The original MOBIL study also showed that the peak magnitude drops when politeness increases from p=0 to p=1 [13]. Figure 7 illustrates the similar shape and the same sensitivity to the politeness factor p, demonstrating consistency between our formal model and empirical observations. However, our rewards define the measurement more precisely. This is because each data point is a reward ratio, specifically $r_{bin} = \mathrm{R}\{\mathrm{LC_bin}\}/\mathrm{R}\{\mathrm{Time_bin}\}$, computed for the closed-loop MDP. This also eliminates sampling and binning artifacts common in simulation studies and establishes a rigorous, reproducible basis for comparing different algorithms or parameter sets.

Our results also align with specific MOBIL findings on parameter influence, going beyond merely reproducing the qualitative shape. MOBIL identifies politeness $p$ as the main driver of lane-change rates, while changes to the incentive threshold $\Delta a_{thr}$ shift the peak only weakly, and the braking limit $\mathrm{b_safe}$ has a negligible effect within realistic ranges. This is because the IDM model already limits decelerations to a comfortable value $\mathrm{b}$ [13]. In our setting, we observe similar effects as increasing $p$ raises the minimum expected number of lane changes and decreases the minimum expected critical-state penalty, as shown in Fig. 5. Reducing $\Delta a_{thr}$ lifts the rate curves but does not alter their qualitative shape. These observations are consistent with the safety-efficiency trade-offs reported by state-of-art [13] and subsequent MOBIL-based analyses. The coupling to IDM is explicit in our model and mirrors the MOBIL practice of computing all accelerations from a longitudinal law . Our formal analysis clarifies a key insight from MOBIL by providing exact reachability statements. When the safety threshold satisfies $\mathrm{b_safe}$ at or below $-4\ m/s^2$, the minimum reachability probability of critical states equals zero, providing a certified boundary on critical-state reachability that holds under worst-case nondeterminism. This formal guarantee complements the empirical observation from simulation studies that stricter braking thresholds enhance safety at the cost of reduced lane-change frequency.

The formal verification methodology outlined in this paper is not intended to reproduce absolute numerical rates from prior studies because of inherent differences in time discretization, state abstraction, and measurement units. Instead, probabilistic model checking allows us to capture and rigorously validate the key behavioral phenomena reported in simulation and learning-based analyses. To our knowledge, this is the first application of probabilistic model checking to the MOBIL algorithm, as prior formal methods either rely on discrete-event model checkers such as Supremica, which provide qualitative defect detection, or apply probabilistic verification to unrelated domains such as parking or multimodal routing. Other approaches based on Signal Temporal Logic evaluate bounded temporal safety but do not yield quantitative probability bounds or expected reward metrics. Because existing formal and simulation-based techniques report fundamentally different types of metrics, direct numerical comparison is not possible, yet the complementarity of these methodologies underscores the contribution of our formally certified guarantees to the safety analysis of autonomous lane-changing systems.

In summary, the results demonstrate that our approach successfully bridges the gap between the realism of simulation and the rigor of formal methods. It preserves the empirically validated traffic-flow characteristics of MOBIL and IDM while augmenting them with verifiable guarantees on safety and performance under uncertainty. This hybrid evidence base, combining realistic trends with certified bounds, offers a more robust foundation for evaluating and deploying autonomous driving strategies.

## 7 Conclusions

In this paper, we present a comprehensive probabilistic model checking framework for the formal analysis of lane-changing algorithms in autonomous vehicles. Our approach systematically formalizes and verifies lane-changing strategies by separating decision rules from traffic dynamics, compiling rules into guards, and analyzing the resulting closed-loop MDP using PCTL and reward queries. This enables precise, density-conditioned measurements of lane-change counts and rates, allowing reproducible comparisons across different models. Using the well-known MOBIL algorithm with IDM as a case study, we identified key parameters influencing lane-change decisions, demonstrated a safety transition where critical states become unreachable for braking limits $b \leq -4$, and quantified trade-offs such as how increased politeness improves deceleration smoothness at a minor cost to time efficiency. Our framework is controller-agnostic and supports fair evaluations across rule-based and learning-based designs. The approach provides valuable insights into the effects of thresholds, parameters, and traffic density on lane-changing behavior. Future research will extend this foundation to address partial observability, richer dynamics, multi-agent interactions, and data-driven calibration, contributing to safer and more reliable autonomous vehicle decision-making.

**Author Contributions** Muhammad Bilal Sarwar and Osman Hasan contributed equally to this work.

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**Materials availability** Not applicable.

**Code availability** The PRISM model code developed during this study is available from the corresponding author upon request.

**Data Availability** Not applicable.

## Declarations

**Competing Interests** The authors have no competing interests to declare that are relevant to the content of this article.

**Ethics Approval and Consent to Participate** Not applicable.

**Consent for Publication** Not applicable.

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